Assignment 2 | Final testing report

Student Course Registration System

SE3002 - Software Quality Engineering

Prepared 12 November 2026 | Requirements-based testing and defect correction

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Verified outcome

78 tests passed; five confirmed defects corrected. The same full suite originally produced 48 passes and 30 failures. Corrected-source coverage is 54/54 statements and 26/26 branches. Original failures and white-box analysis remain preserved separately.

Repository: muhammadaaribali/SQE-Assignment-2

Reading guide

1 Scope and method • 2 Black-box design • 3 Decision table and optimization • 4 Original CFG • 5 Complexity and basis paths • 6 Structural tests and coverage • 7 Condition coverage • 8 MC/DC • 9 Defects and fixes • 10 Collaboration • 11 Evidence and submission.

Assistance and provenance

Codex assisted with test design, implementation, execution, analysis, fixes and reporting at the primary student's request. Account-submitted contributions and assisted reviews are distinguished from independent human work in Section 10. No independent collaborator execution is claimed.

1 | Scope and testing method

The supplied registration.py is the system under test. REQUIREMENTS.md (FR-01 through FR-15) and the matching Student_Registration_SRS.docx are the expected-behavior sources. The assignment requires black-box design, structural testing, GitHub defect management, and this report.

Area	Required behavior
Credits	Current credits 0-18; course credits 1-4; total exactly 18 is allowed.
Prerequisites	Every required course must be completed; an empty requirement set is satisfied.
Registration	Priority: positive dues, duplicate registration, missing prerequisites, credit limit, then availability.
State	Decision calls never mutate records. Successful registration updates membership, credits and enrollment; waitlisting only adds a missing student ID.
Withdrawal	Remove course, subtract credits and decrement enrollment; an unregistered student produces False and no changes.
Validation	Reject negative dues and invalid capacity/enrollment; credit helper validates current credits before new credits.

Three measurements, kept separate

1. Original source + 73 black-box tests: 44 passed, 29 failed.
2. Same original source + five structural additions: 48 passed, 30 failed.
3. Corrected source + the same 78 tests: 78 passed, none failed.

Scope and limitations

Use ordinary integers and correct string/collection types as specified. Databases, authentication, concurrency and automatic waitlist promotion are outside scope. Full measured coverage is not proof of correctness for every possible input.

Black-box expected values were derived from requirements, before structural test design. The assistant had viewed implementation code during the initial explanatory review; this report does not claim complete source blindness. The source remained unchanged through both original-source measurements.

Original baseline: 173a9692f61ce65f801e7799948bd52b9c1ae4b8.
Original source SHA-256: d6c35d6804dd3bcaed3cc164ab2f163ad8214792bdc410e258ec54cdd1099100.

2 | Black-box design: classes and boundaries

The complete design is in docs/blackbox_test_design.xlsx and docs/blackbox_cases.json. It contains 20 ECP and 27 BVA references, exceeding the assignment minima of 8 and 10. This page summarizes the design; each workbook row carries requirement and stable BB test IDs.

Dimension	ECP classes	Boundary values used
Current credits	Valid 0-18; invalid below/above	-1, 0, 1, 17, 18, 19
Course credits	Valid 1-4; invalid below/above	0, 1, 2, 3, 4, 5
Combined credits	Fits vs exceeds maximum	17, 18, 19 (valid individual values)
Dues	Zero, positive, negative	-1, 0, 1
Prerequisites	None required; all, some, none completed	Set membership classes, not numeric BVA
Availability	Seat available; full; invalid occupancy	Capacity 0, 1, 2; enrollment -1, 0, 1, 29, 30, 31 at capacity 30

Representative requirement-based cases

ID / FR	Representative input	Expected
BB-001 / 01,02	check_credit_limit(12, 3)	True
BB-002 / 01,02	check_credit_limit(16, 3)	False
BB-003 / 01,02	check_credit_limit(-1, 1)	Invalid current credits
BB-006 / 01,02	check_credit_limit(12, 5)	Invalid course credits
BB-009 / 03	Completed {CS101}; required {CS101,MATH101}	False
BB-029 / 01,02	check_credit_limit(15, 3)	True (exactly 18)
BB-012 / 04,09,10	Eligible student with dues=1	DUES_BLOCK; unchanged state
BB-014 / 08,10	Eligible student; enrollment=capacity=30	WAITLISTED; unchanged state

Each test changes the targeted input while keeping other fields valid, except intentional priority tests. Invalid numeric cases check exact ValueError messages. A return-type check prevents accepting integer 1/0 in place of Boolean True/False.

3 | Decision table, state and optimization

Five business facts yield 2^5 = 32 combinations. They are recorded in docs/decision_table.xlsx. D = dues positive; R = already registered; P = prerequisites met; C = credits eligible; S = seat available. A dash means irrelevant after a higher-priority outcome, not a runtime measurement.

Rule	D	R	P	C	S	Outcome	Combinations
R1	T	-	-	-	-	DUES_BLOCK	16
R2	F	T	-	-	-	ALREADY_REGISTERED	8
R3	F	F	F	-	-	PREREQUISITES_MISSING	4
R4	F	F	T	F	-	CREDIT_LIMIT	2
R5	F	F	T	T	F	WAITLISTED	1
R6	F	F	T	T	T	REGISTERED	1

Optimization calculation

Design references	Count
ECP / BVA / decision combinations	20 + 27 + 32
Validation-priority / state checks	3 + 11
Total references	93
Exact repeated executions consolidated	20
Unique black-box tests	73

Two designs are duplicates only when operation, complete inputs/state, expected return or exception, and state assertions match. ECP-01 and BVA-10 share BB-001; ECP-03 and BVA-01 share BB-003. Different operations or different later conditions are not duplicates merely because the returned status is identical. All 93 references retain mappings to their 73 executions.

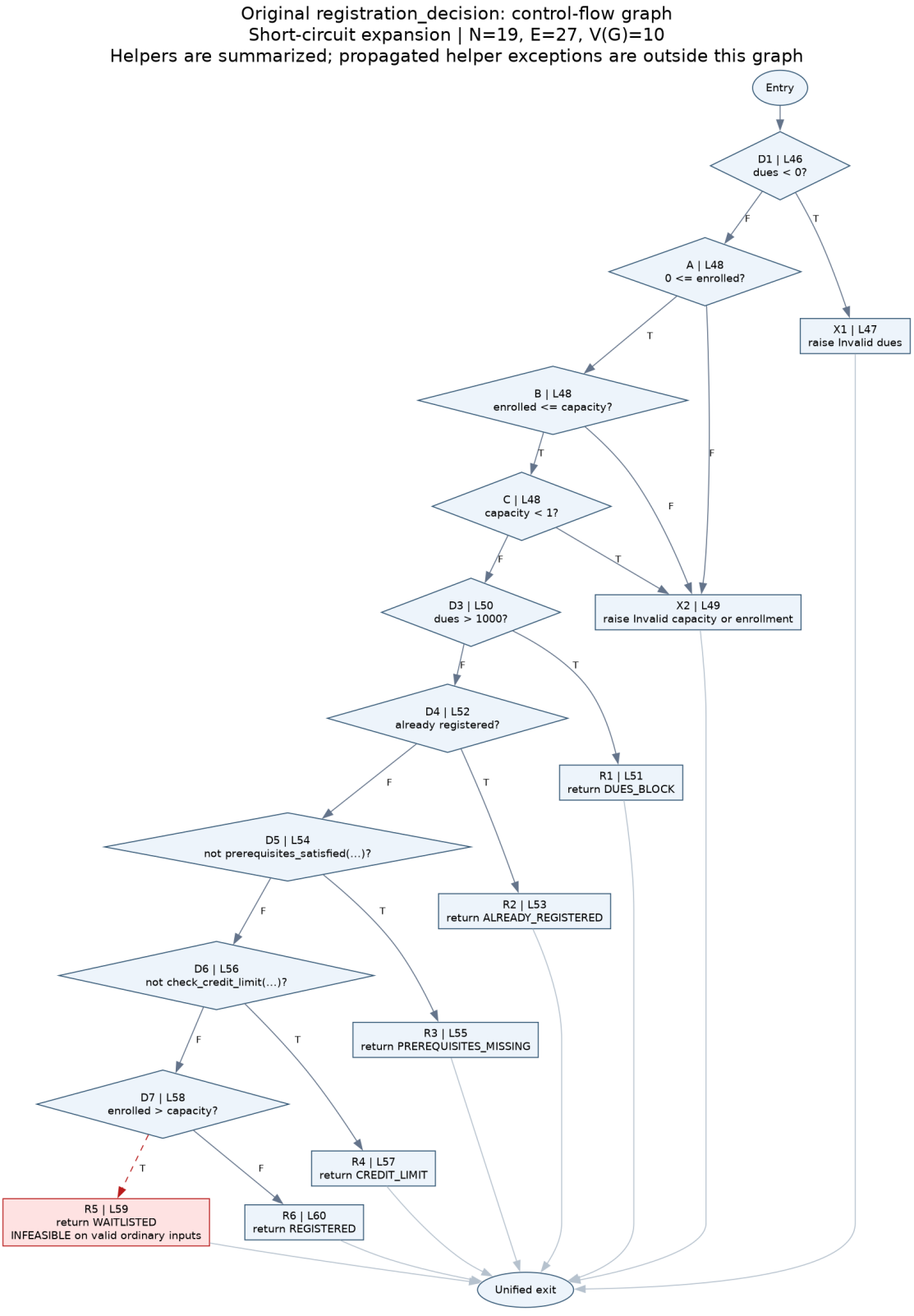
State verification

Fresh dataclass objects and deep copies prevent shared-state leakage. Each decision preserves both complete records. Registration checks the exact membership, credit and enrollment changes. Waitlist tests cover both absent and existing IDs. Rejected operations preserve all fields. BB-072 checks successful withdrawal; BB-073 checks an unregistered withdrawal.

The design JSON and spreadsheets retain their original planning snapshot. Actual execution status is recorded in JUnit/log evidence, not inferred from a historical “Not run” design field.

4 | Original-source control-flow graph

registration_decision only. Diamonds are decisions; T/F arrows are outcomes; rectangles return a result or raise an error. Red marks the infeasible original waitlist route. Helpers are summarized; their internal paths and propagated exceptions are outside this graph.



5 | Complexity and independent basis paths

The graph expands line 48 into three short-circuit predicates: A = 0 <= enrolled, B = enrolled <= capacity, C = capacity < 1. The other six decisions are negative dues, blocking dues, duplicate registration, prerequisites, credit limit and full-course check.

Calculation	Result
Nodes: entry + exit + 9 decisions + 8 return/raise nodes	N = 19
Edges: 1 entry + 18 decision + 8 terminal-to-exit	E = 27
V(G) = E - N + 2 = 27 - 19 + 2	10
Cross-check: binary decision nodes + 1 = 9 + 1	10

If line 48 is instead counted as one compound decision, N=17, E=23 and V=8 (seven source if-statements + 1). This report consistently uses the expanded convention, V=10. These are intraprocedural normal-helper-return counts, not a whole-program/all-exceptions measure.

Path	Node route / exit	Input change	Representative
P1	E,D1,X1,Z	dues=-1	BB-013
P2	E,D1,A,X2,Z	enrolled=-1	BB-020
P3	E,D1,A,B,X2,Z	enrolled=31	BB-019
P4	E,D1,A,B,C,X2,Z	enrolled=0; capacity=0	BB-018
P5	Prefix to D3,R1,Z	dues=1001	WB-001
P6	Prefix to D4,R2,Z	already registered	BB-015
P7	Prefix to D5,R3,Z	completed={ENG101}	WB-003
P8	Prefix to D6,R4,Z	current credits=16	BB-017
P9	Prefix to D7,R5,Z	No feasible original input	None: infeasible
P10	Prefix to D7,R6,Z	Baseline eligible student	BB-011

Baseline: 12 credits, no dues, prerequisites {CS101,MATH101} completed; unregistered for CS201; course 3 credits, capacity 30, enrollment 29. Common prefix: E,D1,A,B,C,D3,D4,D5,D6,D7, stopped at the indicated decision. Each route ends at Z. Ten formal path-edge vectors have rank 10; nine original paths were traced in the audit.

Why P9 cannot execute in the original source

Passing line 48 requires enrollment <= capacity. Neither value changes before line 58 asks enrollment > capacity. Therefore its True outcome is impossible. Above-capacity inputs already raise an error; full courses incorrectly return REGISTERED. The fixed equality check makes this route reachable.

6 | Structural tests and measured coverage

New test	Input and expected behavior	Structural purpose	Original / fixed
WB-001	Dues=1001: DUES_BLOCK	Reach original dues-block line 51	Pass / Pass
WB-002	Dues=1000: DUES_BLOCK	Check wrong implementation cutoff	Fail / Pass
WB-003	Only ENG101 completed: PREREQUISITES_MISSING	Reach missing-prerequisite line 55	Pass / Pass
WB-004	Current credits=-1: exact validation error	Credit-helper error through caller	Pass / Pass
WB-005	Course credits=0: exact validation error	Other helper error through caller	Pass / Pass

All five cases also assert no mutation of either record. WB-004/005 exercise caller-to-helper propagation even though direct helper validation lines were already covered. Tests do not patch the program or manufacture unreachable routes.

Metric	Original BB	Original BB+WB	Corrected BB+WB
Cases / pass / fail	73 / 44 / 29	78 / 48 / 30	78 / 78 / 0
Statements	49/53 = 92.45%	51/53 = 96.23%	54/54 = 100%
Branch outcomes	22/26 = 84.62%	24/26 = 92.31%	26/26 = 100%
Combined coverage.py score	71/79 = 89.87%	75/79 = 94.94%	80/80 = 100%

Coverage gaps and interpretation

Black-box-only missing lines: 51, 55, 59, 71. Missing arcs: 50→51, 54→55, 58→59, 70→71. White-box additions cover the first two lines/arcs. Remaining lines 59 and 71 are blocked by the original waitlist defect. They are retained in coverage denominators.

Before source fixes, white-box additions improve statement coverage by 3.77 percentage points and branch coverage by 7.69 points. The separate corrected-source measurement reaches all statements/branches; the denominator increases to 54 statements because withdrawal now decrements enrollment.

No assertions were weakened. Source/test hashes and JUnit identities verify the same 78 cases: all 30 previous failures become passes and all 48 previous passes remain passing. There are no skips, xfails or exclusions. Condition and MC/DC percentages below are manual, not pytest-cov outputs.

7 | Atomic conditions and evaluation truth table

Scope: nine predicates in the expanded registration_decision graph. Helper return predicates count as one atom each; helper internals are not included in this percentage. NE means not evaluated, never False.

Atom	Original predicate	True witness	False witness
D1	dues < 0	BB-013	BB-011
A	0 <= enrolled	BB-032	BB-020
B	enrolled <= capacity	BB-032	BB-019
C	capacity < 1	BB-018	BB-032
D3	dues > 1000	WB-001	BB-011
D4	already registered	BB-015	BB-011
D5	not prerequisites_satisfied	WB-003	BB-011
D6	not check_credit_limit	BB-017	BB-011
D7	enrolled > capacity	Infeasible	BB-011

Case	D1	A	B	C	D3	D4	D5	D6	D7
BB-013	T	NE	NE	NE	NE	NE	NE	NE	NE
BB-020	F	F	NE	NE	NE	NE	NE	NE	NE
BB-019	F	T	F	NE	NE	NE	NE	NE	NE
BB-018	F	T	T	T	NE	NE	NE	NE	NE
WB-001	F	T	T	F	T	NE	NE	NE	NE
BB-015	F	T	T	F	F	T	NE	NE	NE
WB-003	F	T	T	F	F	F	T	NE	NE
BB-017	F	T	T	F	F	F	F	T	NE
BB-011	F	T	T	F	F	F	F	F	F

Original black-box condition outcomes: 15/18 = 83.33%. With white-box tests: 17/18 = 94.44%. D3=True and D5=True are newly covered; D7=True remains impossible. WB-004/005 raise during D6, so neither supplies a Boolean result for that predicate.

After correction, D3 is dues > 0 and D7 is enrolled == capacity. Existing BB-014/BB-011 demonstrate D7 True/False; the other predicates retain both outcomes. Thus the same nine-predicate manual scope becomes 18/18. This is an analysis of executed witnesses, separate from the instrumented branch measurement.

8 | MC/DC and Python short-circuiting

Select line 48: **D = not (A and B) or C**, where A = 0 <= enrolled, B = enrolled <= capacity, and C = capacity < 1. B is skipped when A is False; C is skipped when not(A and B) is True.

Row / case	enrolled, capacity	Logical A,B,C	Evaluated A,B,C	D
M0 / BB-032	0,30	T,T,F	T,T,F	F
M1 / BB-020	-1,30	F,T,F	F,NE,NE	T
M2 / BB-019	31,30	T,F,F	T,F,NE	T
M3 / BB-018	0,0	T,T,T	T,T,T	T

Target	Pair	Independent effect
A	M0 / M1	B logically T and C logically F stay fixed. A changes and D changes; B/C are masked in M1.
B	M0 / M2	A=T and C logically F stay fixed. B changes and D changes; C is masked in M2.
C	M0 / M3	A=T and B=T stay fixed. C changes and D changes; all atoms execute.

Short-circuit-aware masking MC/DC: 3/3 conditions = 100% for this selected decision. All four witness cases already belonged to the original black-box suite, so this selected MC/DC value is 100% before and after structural additions. The validation expression is unchanged by source fixes.

Logical truth values for skipped pure comparisons can be derived from the unchanged integer inputs, but are not counted as executed outcomes. Under a stricter convention requiring every non-target atom to execute in both pair members, A/B pairs cannot satisfy that convention. The masking convention is explicit.

Other compound expressions

Credit helper line 30 checks 0 <= current <= 18: BB-003, BB-004 and BB-021 cover F/NE, T/F and T/T. Line 32 checks 1 <= new <= 4: BB-005, BB-006 and BB-025 cover the same patterns after valid current credits.

register_course line 70 checks WAITLISTED and ID absent. In the original program the first condition is False and the second is NE. After fixes BB-064/065 cover absent/present IDs. No full-program MC/DC percentage is claimed.

Method references: coverage.py measurement; NASA MC/DC tutorial.

9 | Defects, reproductions and verified fixes

Five distinct defects explain multiple failing cases; 30 failures are not 30 separate defects. All five Issues are now closed by merged PR #9. The issue bodies retain the original observations and include verified resolution notes.

Issue / FR	Reproduction and original actual	Expected / correction	Witness
#3 / FR-01	check_credit_limit(15,3) returned False	True; replace <18 with <=18	BB-029
#4 / FR-03,06	Completed {CS101}, required {CS101,MATH101}: returned True	False; required.issubset(completed)	BB-009,016
#5 / FR-04,09	Eligible student owes Rs.1: REGISTERED	DUES_BLOCK; dues >0	BB-012, WB-002
#6 / FR-08,12	Eligible student; capacity=30, enrolled=30: REGISTERED; applying decision raises enrollment to 31	WAITLISTED; use enrolled == capacity; preserve other fields	BB-014,064,065
#1 / FR-13	Withdraw registered 3-credit course: credits 15→12 but enrollment stays 30	Enrollment 30→29; decrement once	BB-072

Regression and status

The five source corrections are isolated in the defect-fixes branch and PR #9. Merged commit: 1a9e7f4c3f528e5818171c69546be353f0987284. Both local and GitHub runs pass the unchanged 78 cases, with no missing statements or branches.

Input validation order, business-decision priority, signatures, return types and error/status strings remain unchanged. BB-073 still proves an unsuccessful withdrawal does not mutate either record. Original artifacts and flowchart are not overwritten with corrected-source results.

Issues: #1 | #3 | #4 | #5 | #6

Interpretation

The corrected equality check makes waitlisting reachable. The original flowchart remains evidence for Part B; do not interpret its old >1000 and >capacity predicates as current corrected logic. The corrected function has the same expanded topology and complexity V=10.

10 | Collaboration and contribution record

Member	Scope and observed record
Muhammad Aarib Ali 24P-3036 muhammadaaribali	Primary owner: test design, 71 primary black-box cases, structural analysis, defect investigation/fixes, automation and report. Directed the staged work, discussed case/PR/CFG concepts, supplied identities and authorized actions. Codex prepared and executed technical artifacts under this direction.
Shafay Naeem 3089 shafaynaeem	Assigned scope: withdrawal FR-13/14 (BB-072/073). Codex implemented and executed these and submitted through the signed-in Shafay account at the primary student's request. Assisted review on PR #7 was submitted after the primary student stated Shafay understood. Independent human implementation/execution is not claimed.

PR	Contribution / branch	Review and merge record
#2	Withdrawal tests; collaborator-withdrawal-tests → blackbox-testing	Primary-account assisted approval 5857526324; merged 8e1f658.
#7	Full black-box execution; primary-blackbox-execution → main	Shafay-account assisted comment 5859018400 and approval 5859050351; merged 6b29e0c.
#8	White-box analysis; whitebox-testing → main	Codex-assisted technical COMMENT 5878893248; merged e72449f.
#9	Five corrections; defect-fixes → main	Codex-assisted technical COMMENT 5879011434; merged 1a9e7f4.

PR links: #2 | #7 | #8 | #9

Review interpretation and viva

The assignment asks every member to review another member's PR and explain individual work. Cross-account review records exist, but they disclose AI assistance and are not evidence of independent human review or execution. Technical checks found no blocking findings on PRs #8/#9; no reviewer feedback or human contribution has been invented. Both students should personally inspect their assigned work and be prepared to explain it in the viva.

11 | Evidence index and submission checks

Measurement	GitHub Actions run	Outcome / durable files
Original black-box	42997171307	44 pass / 29 fail; docs/evidence/blackbox-ci*
Original full suite	43036997090	48 pass / 30 fail; docs/evidence/whitebox-ci*
Corrected full suite	43268281508	78 pass / 0 fail; docs/evidence/fixed-ci*
Fixes final head	43268576633 / 43268561611	Both successful; code/tests unchanged from the preserved corrected run

Actions run 42997171307

Actions run 43036997090

Actions run 43268281508

The workflow installs requirements, records Python/dependencies/commit, runs pytest with statement/branch coverage, and uploads logs, JUnit and coverage artifacts even on failure. Genuine failures keep a failed job status. CI uses Python 3.14.10 for the preserved Stage 4/5 runs; local recorded runs use Python 3.14.4.

Required artifacts

Assignment area	Repository evidence
A1-A3 / A5	docs/blackbox_test_design.xlsx; docs/decision_table.xlsx; docs/blackbox_cases.json
A4 / B4	tests/test_blackbox.py; tests/test_whitebox.py
B1-B3	docs/cfg.png; docs/WHITEBOX_ANALYSIS.md; evidence/whitebox-analysis-check.json
C1-C5	Feature-branch history, linked Issues/PRs, .github/workflows/tests.yml
D	docs/testing_report.pdf (this report); docs/DEFECT_FIXES.md
Original source / corrected source	Baseline 173a969; original full suite e72449f; fixes merged 1a9e7f4

Reproduce from the repository root

```
python -m pip install -r requirements.txt
python -m pytest tests/ -v --tb=short --cov=registration --cov-branch --cov-report=term-missing
```

Expected on corrected main: 78 passed, exit 0. For original measurements use their preserved commit and matching suite; do not replace historical failures. Submit the repository link, this report, design workbooks and referenced evidence according to the course portal instructions. Portal submission and instructor access are not asserted by this report.